

An Adaptive Digital Image Processing Pipeline for Smart-City Surveillance under Noise, Low Light, Interference and Transmission Loss

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Abstract—Islamabad’s public-space surveillance network suffers from sensor noise, under-exposed night frames, electrical interference, motion blur and packet loss, all of which degrade the identification of pedestrians and vehicles. We present a complete image-processing pipeline that treats these faults as an inverse problem rather than a fixed filter chain: five blind diagnostics measure each frame, and only the restoration stages the diagnosis actually calls for are executed, with a user-selected *profile* capping how expensive each stage may be. The system implements six technical tasks—spatial filtering, frequency-domain filtering, restoration and reconstruction, edge- and region-based segmentation, JPEG and wavelet compression, and object recognition—with the core operators written from first principles and verified against reference libraries. Every design trade-off is measured rather than asserted. Blind diagnosis flags Gaussian noise, impulse noise, periodic interference, night exposure and packet loss in 100 % of degraded frames at a false-alarm rate of at most 5 % on clean frames. Adaptive restoration raises pedestrian F_1 from 0.624 to 0.917 under 20 % salt-and-pepper noise and from 0.509 to 0.733 on compound night frames. Marker-controlled watershed segmentation attains IoU 0.692 at 3.5ms per frame, beating GrabCut on both accuracy and cost. Region-of-interest coding gains 7.0 dB on people at equal file size in a wide CCTV view, and pixelating heads preserves analytics (F_1 0.877 versus 0.900) while removing identities. Evaluation on real CCTV shows detection is limited by object scale rather than image quality. The profiles span real-time and quality regimes on one CPU with no GPU.

Index Terms—Digital image processing, video surveillance, image restoration, image segmentation, image compression, object detection, privacy protection, real-time systems.

I. INTRODUCTION

THE city of Islamabad has deployed a smart surveillance system for real-time monitoring of public spaces. In practice the deployment is limited less by the analytics than by the quality of the imagery reaching them: cameras operate at night with short exposure and high sensor gain, share infrastructure with power electronics that inject periodic interference, observe fast-moving vehicles that motion-blur, and stream over wireless links that drop packets. The result is that pedestrians and vehicles which a detector would find easily in clean imagery are missed.

A conventional response is a fixed enhancement chain—denoise, sharpen, brighten—applied to every frame. This is wasteful, because a clean daytime frame pays for filters it

does not need on a node whose compute budget is shared across many cameras; and harmful, because enhancement operators are not benign: a median filter erases fine detail, non-local means removes the weak edges edge-based segmentation depends on, and histogram equalisation amplifies exactly the noise a dark frame contains most of.

This work therefore treats frame enhancement as a *decision* problem. The system measures the frame—noise standard deviation, impulse density, spectral peak prominence, mean luminance, dark-region fraction and lost-block ratio—then executes only the operators those measurements justify. A profile (fast, balanced, quality) independently caps the cost of each selected operator, converting the real-time-versus-quality requirement from a discussion point into a run-time parameter.

A. Contributions

- 1) A *blind diagnostic front end* calibrated against measured false-alarm rates on clean imagery, establishing which faults are separable from frame statistics alone—and which are not.
- 2) An *adaptive restoration pipeline* whose stage ordering is justified experimentally rather than by convention, including two non-obvious orderings: repairing pixel-level defects before any resampling, and judging noise by its post-brightening magnitude.
- 3) *From-scratch implementations* of the core operators—convolution, median and adaptive median filters, the DFT filter family, Otsu, histogram equalisation, region growing, an 8×8 DCT codec and a wavelet codec—each verified against OpenCV or scikit-image.
- 4) A *measured account of every design trade-off*, with the negative results retained: uneven illumination is not blindly detectable, denoising before segmentation does not help, and the largest detector input size is not the most accurate.
- 5) An explicit *module interdependence* analysis quantifying what each upstream stage is worth to the downstream task.

II. PROBLEM ANALYSIS AND CONSTRAINTS

A. Fault taxonomy and the degradation model

Public surveillance datasets contain reasonably clean imagery, whereas PSNR and SSIM require a known clean reference that real night footage cannot provide. We therefore adopt the standard linear degradation model [1]

$$g(x, y) = H[f(x, y)] + \eta(x, y), \quad (1)$$

where f is the clean frame, H a (possibly identity) degradation operator and η an additive noise field, and synthesise each fault with documented, seeded parameters. Seven faults are modelled: Gaussian sensor noise, salt-and-pepper bit errors, sinusoidal mains interference, linear motion blur, under-exposure, a multiplicative illumination field, and zeroed 16×16 blocks standing for dropped packets. A compound “worst night” scenario combines under-exposure, sensor noise and sparse bit errors, because real failures co-occur.

B. Constraints

Three constraints shape every design decision that follows.

Real-time performance versus processing quality. A node must keep up with its camera. Restoration cost is dominated by a few expensive operators (non-local means, per-frame FFT, large detector inputs), so the system must be able to trade them away rather than being compiled around one fixed choice.

Resource constraints versus performance. The target is an edge-class device without a GPU, which rules out large detectors, motivates YOLOv4-tiny through OpenCV’s DNN module (no PyTorch, no CUDA), and makes peak memory a reported metric.

Security and privacy versus system accessibility. A stream that exposes identities is a liability, but one that destroys them also destroys the analytics justifying the system. The design must hide identity while preserving the evidence counting and tracking depend on, and let an authorised investigator recover the original.

III. SYSTEM DESIGN: PIPELINE AND FLOWCHARTS

A. Pipeline overview

Fig. 1 shows the eight stages: the frame is diagnosed and pixel-level defects repaired at native resolution; the profile rescales it; restoration removes interference, noise and lighting defects; a detector locates people and vehicles; segmentation refines those boxes into silhouettes; a privacy stage hides identities; and a region-of-interest codec spends bits where the objects are. Only the selected stages execute.

B. Adaptive decision logic

Fig. 2 is the decision flowchart actually implemented in `surveillance/pipeline.py`; the thresholds printed on it are read from the source at drawing time, so the figure cannot drift from the code. Table I lists each threshold together with the measured separation that justifies it.

TABLE I
DECISION THRESHOLDS AND THE MEASURED SEPARATION JUSTIFYING EACH (SECTION VI-A).

Statistic	Threshold	Measured separation
impulse ratio	0.004	clean $\leq$ 0.001; 5% S&P $\geq$ 0.033
noise $\hat{\sigma}$	7.5	clean 0.6–8.1 (95th pct. 5.8); $\sigma=15 \Rightarrow \geq 9.4$
mean luminance	50	clean $\geq$ 62; simulated night 12–34
dark-region frac.	0.30	not separable — see Section IX
lost-block ratio	0.002	clean 0; 3% loss $\approx$ 0.028
spectral prominence	2.0	clean $<$ 2.0; amplitude-40 interference 1.6–4.3

TABLE II
THE THREE OPERATING PROFILES.

	fast	balanced	quality
resolution	$0.5 \times$	$0.75 \times$	$1.0 \times$
impulse filter	median 3×3	median 3×3	adaptive median
denoiser	Gaussian	bilateral	NLM + unsharp
periodic notch	—	yes	yes
illumination	gamma	gamma+CLAHE	+homomorphic
detector input	320, every 3rd	416	608
segmentation	MOG2 masks	watershed	watershed
JPEG q (ROI/bg)	85 / 25	90 / 30	92 / 40

C. Profiles

A profile is one object fixing the working resolution and the cost ceiling of each stage (Table II). It never changes *whether* an operator runs—that remains the diagnosis’s decision—only *how expensive* it may be.

D. Three deliberate orderings

(i) *Repair before resize.* Area-interpolated downsampling averages an isolated saturated pixel into its neighbourhood, converting a detectable impulse into an undetectable local bias; the same applies to a zeroed 16×16 block. Both are therefore repaired at native resolution.

(ii) *Denoise before brightening, judged after gain.* Low-light enhancement multiplies intensities by an effective gain $\gamma \approx \text{clip}(110/\bar{L}, 1, 4)$, so noise invisible in a dark frame is amplified into visibility. The system tests $\hat{\sigma} \cdot \gamma$ rather than $\hat{\sigma}$, and denoises *before* enhancing.

(iii) *Luminance before uniformity.* A uniformly dark frame and an unevenly lit one need opposite corrections, so only a frame bright *on average* is considered for homomorphic correction.

(iv) *Re-estimate after repair.* Noise is re-estimated after impulse and notch removal, since both change the residual the denoiser must match.

IV. METHODS

This section maps the six technical tasks onto the implementation. Operators marked (*scratch*) are written from first principles and verified in Section V.

A. Task 1: Spatial filtering

Two-dimensional correlation (*scratch*) with reflect-101 borders underpins the mean and Laplacian filters, and the median is implemented (*scratch*) with sliding-window order statistics.

Adaptive surveillance pipeline — only the stages the diagnosis calls for are executed

a profile (fast / balanced / quality) caps how expensive each selected stage may be

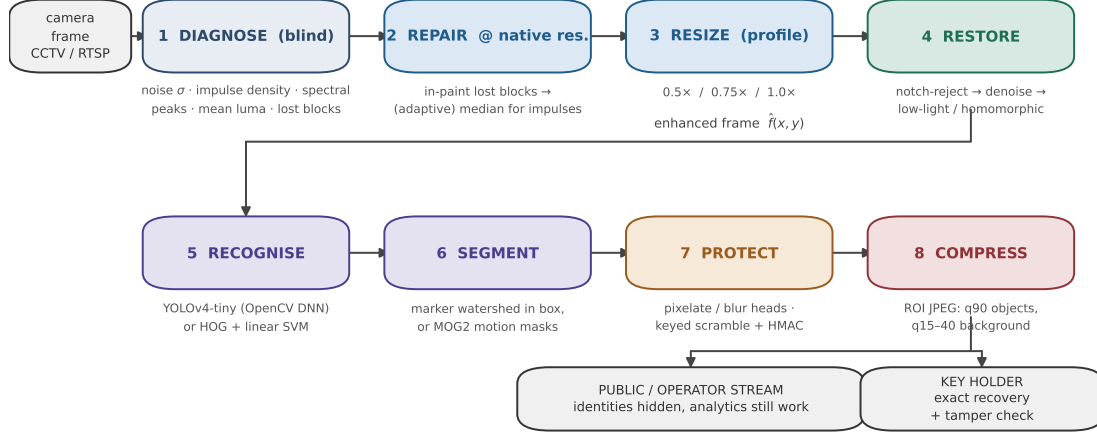


Fig. 1. End-to-end system pipeline. Stages 1–2 run at native resolution because area resampling would average impulses and lost blocks into blobs that no later filter can identify; the profile then fixes the working resolution and the cost ceiling of every subsequent stage. Detection boxes are the common currency of stages 6–8: they tell segmentation *where* to look, privacy *what* to hide and the codec *which* regions deserve bits.

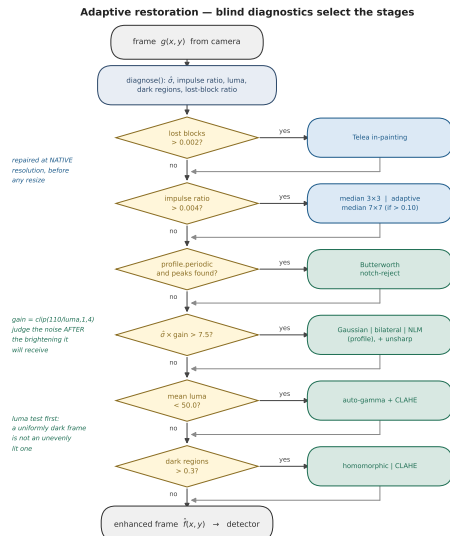


Fig. 2. Adaptive restoration flowchart. Each diamond is a blind test on the diagnostics; the “yes” branch executes an operator whose strength is set by the profile, and control rejoins the main path. Three orderings are deliberate and are justified in Section III-D.

The *adaptive median* [1] grows its window until the local median is not itself an impulse ($z_{\min} < z_{\text{med}} < z_{\max}$) and replaces a pixel only when that pixel is extreme, preserving detail where a fixed median smears it. Edge-preserving smoothing uses the bilateral filter [2] with σ_{color} tied to the estimated noise, and non-local means [3] in the quality profile.

B. Task 2: Frequency-domain filtering

The ideal, Butterworth and Gaussian low- and high-pass transfer functions $H(u, v)$ are built on a centred distance grid and applied as $g = \mathcal{F}^{-1}\{H \cdot \mathcal{F}\{f\}\}$ with reflect padding, with

cut-offs expressed as a fraction of image size for resolution independence.

The interference remover is automatic: the Hann-windowed log-magnitude spectrum is scored for spikes by subtracting, at each frequency, the mean of an annulus $4 \leq r \leq 9$ bins away from the mean of the 3×3 core—a pure sinusoid concentrates into the core while the $1/f$ scene background is flat across both—and conjugate-symmetric peaks above a prominence of 2.0 are suppressed with an order-4 Butterworth notch-reject product filter. Uneven lighting is handled homomorphically: since $f = i \cdot r$, logarithms separate illumination from reflectance, so attenuating low frequencies ($\gamma_L=0.4$) while preserving high ones ($\gamma_H=1.1$) flattens the illumination field without flattening the scene.

C. Task 3: Restoration and reconstruction

Deconvolution is implemented as the inverse filter $\hat{F} = G/H$, the Wiener filter $\hat{F} = [H^*/(|H|^2 + K)] G$, and constrained least squares with a Laplacian penalty. Low-light restoration offers auto-gamma, global histogram equalisation (*scratch*) and CLAHE [4]; lost blocks are located by an all-black block test and filled by Telea in-painting [5].

Blind diagnostics use an edge-aware Immerkær estimator [6]: the difference-of-Laplacians response is summed only over the 70 % of pixels with the weakest Sobel response [7], because the plain estimator over-reads on textured streets.

D. Task 4: Segmentation

Both families are implemented and compared inside detector boxes, which is how a surveillance system uses segmentation: the detector says *where*, the segmenter says *which pixels*. Edge-based methods use Canny [8] or Sobel gradients

followed by morphological closing, hole filling and largest-component selection. Region-based methods are Otsu (*scratch*) [9] on a colour-distance map, seeded region growing (*scratch*), marker-controlled watershed and GrabCut [10]. For static cameras a motion segmenter combines MOG2 [11]—or a from-scratch running average—with morphology and connected components.

E. Task 5: Compression

Baseline JPEG through libjpeg is the production reference. A from-scratch JPEG-like codec performs colour transform, 4:2:0 subsampling, 8×8 DCT, IJG-scaled quantisation, zig-zag scan, DC differential coding and entropy coding, exposing every stage; a wavelet codec [12] applies a multi-level DWT with dead-zone quantisation. All three report *measured* compressed byte counts, so bits per pixel is observed rather than estimated. Region-of-interest coding encodes detected objects at high quality and the background at low quality, resolving the compression–fidelity trade-off per region instead of per frame.

F. Task 6: Object recognition

Two detectors share one interface: HOG with a linear SVM [13] as the classical baseline, and YOLOv4-tiny [14] through OpenCV’s DNN module, detecting people and vehicles at three input sizes on a CPU. Greedy non-maximum suppression is implemented (*scratch*).

G. Privacy and evidence integrity

Three tiers are provided [15]: irreversible anonymisation (pixelating or blurring heads, bodies or silhouettes) for the public stream; a keyed scrambler permuting 8×8 blocks under a key-derived keystream, from which a key holder recovers the original bit-exactly; and an HMAC-SHA256 tag [16] per stored frame so tampering is detectable.

V. EXPERIMENTAL SETUP

A. Datasets

Table III lists the data used. Penn-Fudan [17] is the primary still-image set because it is the only freely available pedestrian set with *pixel-accurate instance masks*, which IoU and Dice require and bounding-box datasets cannot supply; its clean frames also serve as the references (1) demands. CAVIAR [18] supplies real CCTV with hand-labelled boxes and a KITTI [19] clip supplies vehicles.

Three candidate datasets are deliberately not used. Cityscapes [20] requires registered access and roughly 12 GB, conflicting with the requirement that every result be regenerable from one unattended download. The Lahore Urban Traffic Observations set is tabular sensor data, not imagery. A face-mask dataset addresses mask classification on tightly cropped faces—orthogonal to the six tasks, and exercising neither restoration nor compression under realistic surveillance geometry.

TABLE III

DATA USED. ALL SOURCES DOWNLOAD WITHOUT REGISTRATION AND ARE VERIFIED ON ARRIVAL (FILE COUNTS, MD5, XML PARSE).

Dataset	Content	Used for
Penn-Fudan [17]	170 images, 423 mask instances	PSNR/SSIM references, IoU/Dice, $P/R/F_1$
CAVIAR [18]	5 sequences, 2211 frames, CVML boxes	real-CCTV detection, motion segmentation
KITTI clip [19] vtest.avi	16.6 s street video 768×576, static camera	vehicle robustness real-time benchmark, ROI coding
YOLOv4-tiny [14]	COCO weights, 24 MB	recognition

TABLE IV

TRIGGER MATRIX (% OF 60 FRAMES ON WHICH EACH DIAGNOSTIC FIRES). DIAGONAL ENTRIES ARE CORRECT DETECTIONS; THE *clean* ROW IS THE FALSE-ALARM RATE.

Scenario	imp.	noise	per.	dark	d.reg	lost
clean	0	2	5	0	0	0
Gaussian $\sigma=15$	0	100	5	0	0	0
Gaussian $\sigma=30$	0	100	5	0	0	0
salt & pepper 5 %	100	100	5	0	0	0
salt & pepper 20 %	100	100	0	0	0	0
periodic	0	2	100	0	0	0
motion blur	0	0	5	0	0	0
low light	0	0	3	100	0	0
uneven light	0	0	5	2	5	2
block loss	0	2	2	0	2	100
night compound	100	53	5	100	0	0

B. Metrics

Image quality uses MSE, PSNR and SSIM [21]; segmentation uses IoU and Dice; detection uses greedy score-ordered one-to-one matching at IoU 0.5 with precision, recall, F_1 and all-point interpolated AP [22]. Because Penn-Fudan leaves small and heavily occluded people unlabelled, unmatched detections shorter than 90 px are ignored rather than counted as false positives, identically for every detector.

C. Verification and reproducibility

Every from-scratch operator is checked against an independent reference in a 17-test suite: convolution against `cv2.filter2D` to 10^{-8} ; mean and median filters against OpenCV exactly; Otsu and histogram equalisation to within one grey level; PSNR and SSIM against scikit-image; and the scrambler against exact round-trip and key-sensitivity properties. All randomness is seeded, so the result set regenerates from one command. Experiments ran single-process on an Apple M-series CPU with Python 3.14, OpenCV 4.14 and NumPy 2.5.

VI. RESULTS AND ANALYSIS

A. Can each fault be detected blindly?

Before any filter can be selected adaptively, the system must be able to recognise what is wrong with a frame using only the frame itself. Table IV is the resulting trigger matrix over 60 images: each row is a scenario, each column the percentage of frames on which a given diagnostic fired.

Five of the six fault classes are detected on 100 % of degraded frames while firing on at most 5 % of clean frames.

TABLE V
SPATIAL FILTERING (40 IMAGES). BEST FILTER PER SCENARIO,
COMPARED WITH THE CHEAPEST COMPETITIVE ALTERNATIVE.

Scenario	Filter	PSNR	SSIM	ms
Gaussian $\sigma=15$	(degraded)	24.77	0.591	—
	NLM	28.51	0.795	208.6
	Gaussian	27.35	0.792	0.3
Gaussian $\sigma=30$	(degraded)	19.02	0.362	—
	Gaussian	24.47	0.676	0.4
	NLM	24.28	0.660	208.5
S&P 5 %	(degraded)	17.93	0.448	—
	adaptive median	31.83	0.945	15.5
	median 3×3	27.82	0.851	0.2
	bilateral	17.91	0.430	1.4
S&P 20 %	(degraded)	11.92	0.153	—
	adaptive median	29.42	0.925	17.1
	median 3×3	24.97	0.795	0.1

The compound night scenario triggers both the impulse and darkness rules; its noise rule fires on only 53 % of frames, an under-detection the gain-aware test of Section III-D compensates, since a frame flagged as dark has its noise re-evaluated after the brightening gain.

The noise estimator is monotonic in true σ but systematically under-reads it—10.7 at $\sigma=15$, 19.5 at $\sigma=30$, 25.0 at $\sigma=40$ —because clipping at 0 and 255 discards noise energy and the homogeneous-region restriction samples the flattest 70 % of the image. This is acceptable because the threshold was calibrated against the estimator’s measured output (clean frames average 2.4, maximum 8.7; $\sigma=15$ yields a minimum of 9.8) rather than nominal σ : separation at the decision boundary, not absolute accuracy, is what the pipeline requires.

Periodic interference detection rises with amplitude: 5 % at amplitude 0 (the false-alarm floor, caused by periodic scene content such as railings), 81.7 % at 10, 98.3 % at 20 and 100 % at 40 and above.

The one clear negative result is uneven illumination, whose diagnostic fires on only 5 % of frames—indistinguishable from the clean false-alarm rate. A clean street scene with ordinary shadows produces the same dark-region statistic as genuinely non-uniform lighting, because the statistic cannot tell a dark *object* from a dark *illuminant*. Rather than tune the threshold until it appears to work, we report it as unusable and expose homomorphic correction as a per-camera setting (`illumination="always"`).

B. Task 1: Spatial filtering

Table V reports the best filter per scenario together with the cost of obtaining it, using the *blind* $\hat{\sigma}$ estimate to set filter strength, exactly as the deployed system would.

The *matched* filter matters far more than the expensive one. On 20 % impulse noise the adaptive median gains 17.5 dB over the degraded input and 4.4 dB over a fixed 3×3 median, whereas non-local means—the most expensive filter available, at 208 ms—reaches only 16.34 dB, some 13 dB worse, because an averaging filter cannot repair saturated outliers. Bilateral filtering is worse still (12.12 dB), since an impulse is precisely the large intensity difference its range kernel is designed to

TABLE VI
REMOVING PERIODIC INTERFERENCE (30 IMAGES, RANDOM
INTERFERENCE FREQUENCIES).

Method	PSNR	SSIM	ms
(degraded)	22.30	0.530	—
spatial mean 5×5	23.07	0.546	0.2
spatial median 5×5	23.22	0.539	0.3
ideal LPF $D_0=0.10$	22.01	0.512	44.1
Butterworth LPF $D_0=0.10$	23.12	0.594	49.9
Gaussian LPF $D_0=0.10$	23.35	0.592	46.1
auto notch-reject	34.78	0.943	94.9

preserve. Choosing the wrong filter family costs far more than choosing a cheap member of the right one—the entire justification for diagnosing before filtering.

The cost/benefit picture for Gaussian noise is equally pointed. At $\sigma=15$ non-local means is best (28.51 dB) but takes 208.6 ms, while a simple Gaussian blur reaches 27.35 dB—96 % of the SSIM—in 0.3 ms, a $700 \times$ speed-up for 1.2 dB. At $\sigma=30$ the Gaussian filter *overtakes* NLM outright (24.47 versus 24.28 dB). This is why the fast profile’s cheap Gaussian denoiser is not merely a compromise, and why NLM is confined to the quality profile.

Sharpening after denoising is counter-productive in every scenario tested: `median3+laplacian` and `nlm+unsharp` always score below the corresponding filter alone, because both amplify the residual noise the denoiser left behind. Sharpening is consequently gated behind the quality profile and applied only at low amplitude.

1) *From-scratch implementations versus OpenCV*: The scratch NumPy implementations are numerically *identical* to OpenCV’s (maximum absolute difference 0 for all four operators) while being 5–305 $\times$ slower: mean 5×5 , 8.16 versus 0.07 ms; median 3×3 , 19.89 versus 0.07 ms (305 $\times$); median 5×5 , 65.05 versus 0.23 ms; Laplacian, 4.11 versus 0.79 ms (5.2 $\times$). The scratch versions demonstrate the mechanism and validate the library; the library versions serve the real-time path. The Laplacian’s smaller gap reflects a 3×3 kernel dominated by memory traffic rather than the arithmetic SIMD accelerates in the median’s sorting network.

C. Task 2: Frequency-domain filtering

1) *Periodic interference*: Table VI compares the automatic notch-reject filter against global low-pass filtering and spatial smoothing on frames corrupted with randomly placed sinusoidal interference.

The notch filter gains **12.5 dB** over the degraded input against at most 1.1 dB for the best global low-pass—an order-of-magnitude difference that illustrates the central argument for frequency-domain processing. Periodic interference is spatially delocalised but spectrally compact: it occupies two conjugate points in $F(u, v)$ and everywhere in $f(x, y)$, so a spatial smoother must destroy the signal to attenuate it, whereas a notch removes it almost exactly. Fig. 3 shows the detected peaks, the filter and the restored frame.

2) *The low-pass filter family and ringing*: On Gaussian noise the three transfer functions rank consistently Gaussian > Butterworth > ideal at every cut-off: at $D_0=0.10$ they give

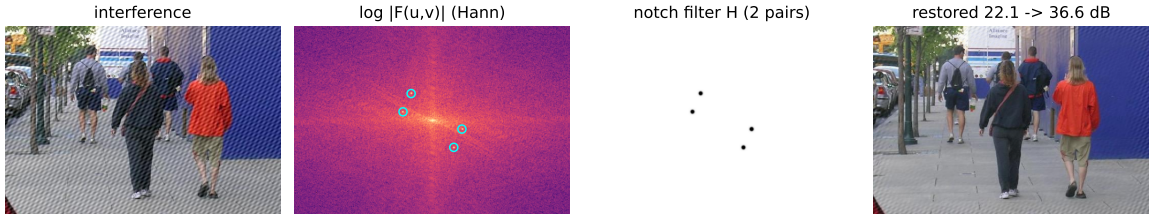


Fig. 3. Automatic notch rejection. Left to right: interfered frame; its Hann-windowed log-magnitude spectrum with the detected conjugate peak pairs circled; the resulting Butterworth notch-reject filter $H(u, v)$; and the restored frame.

24.07, 23.59 and 22.79 dB respectively. The ideal filter is worst despite being the most aggressive, because its discontinuous transfer function has a sinc impulse response that produces Gibbs ringing. Notably, at the tightest cut-off ($D_0=0.05$) the ideal filter is *worse than no filtering at all* in PSNR (20.33 versus 20.49 dB): the ringing artefacts it introduces cost more than the noise it removes. This is the classical textbook result [1] reproduced on real surveillance imagery, and it is why every low-pass in the deployed pipeline is Gaussian.

3) *Uneven illumination*: Homomorphic filtering raises SSIM from 0.902 to **0.922** and PSNR from 16.83 to 18.17 dB. CLAHE reaches a similar PSNR (18.03 dB) but a *lower* SSIM (0.859), and global histogram equalisation is worse than doing nothing on both (16.37 dB, 0.849). The PSNR/SSIM divergence is instructive: CLAHE improves average intensity error while damaging local structure, whereas the homomorphic filter is the only method operating on the correct model—since $f = i \cdot r$ is multiplicative, logarithms make it additive, so attenuating low frequencies removes the illumination field rather than re-scaling the histogram. A percentile stretch instead of the mean-restoring output mapping lowers both metrics (14.84 dB, 0.884).

4) *Frequency versus spatial implementation*: Filtering in the two domains is equivalent to within numerical precision: a Gaussian applied as $H(u, v)$ and as a spatial kernel agree to 56.6–59.0 dB PSNR, which is round-off. The cost, however, is not equivalent. For $\sigma=1$ px the spatial convolution takes 0.29 ms against 84.2 ms for the DFT route ($290\times$); even at $\sigma=8$ px the spatial form remains $24\times$ faster (3.6 versus 86.5 ms), because the DFT cost is fixed by image size while the spatial cost grows only with kernel support. The frequency domain is therefore reserved in this system for operations with no compact spatial equivalent—notch rejection and homomorphic filtering—rather than used for ordinary smoothing.

D. Task 3: Restoration and reconstruction

1) *Motion deblurring*: Table VII compares the three deconvolution filters on frames blurred by a 15 px horizontal motion PSF with additive noise.

The inverse filter is catastrophic: 6.67 dB, i.e. 15.8 dB *worse* than the blurred input it was meant to repair. This is the textbook failure made concrete. Where $H(u, v) \rightarrow 0$ the quotient $G/H = F + N/H$ is dominated by N/H , so the noise is amplified without bound. The Wiener filter differs

TABLE VII
DECONVOLUTION OF A 15 PX MOTION BLUR ($\sigma=2$ NOISE), 30 IMAGES.

Method	PSNR	SSIM	ms
(degraded)	22.49	0.653	—
inverse filter	6.67	0.027	50.4
Wiener $K=0.001$	19.68	0.393	50.9
Wiener $K=0.005$	24.27	0.586	50.4
Wiener $K=0.01$	25.52	0.673	50.3
Wiener $K=0.03$	25.20	0.751	50.1
CLS $\gamma=0.02$	22.42	0.664	66.8
Laplacian sharpen (no PSF)	19.02	0.390	3.5

only by the additive term K in the denominator, yet that term is the difference between a 6.67 dB result and a 25.52 dB one. Its optimum is broad— K between 0.005 and 0.03 all work—which matters in practice because K is a noise-to-signal ratio that must itself be estimated. Note also that PSNR and SSIM disagree about the optimum ($K=0.01$ versus $K=0.03$): the larger K suppresses more noise at the cost of residual blur, which SSIM rewards.

The PSF, however, must be known. With the correct 15 px length, Wiener yields 24.27 dB; assuming 12 px gives 22.16 dB and assuming 18 px gives 19.63 dB. A $\pm 20\%$ error therefore costs 2.1–4.6 dB, and at 21 px (a 40% over-estimate) the output at 17.86 dB is *worse than the blurred input*. Over-estimation is more damaging than under-estimation because too long a PSF places spurious zeros inside the signal band. Since no blind PSF estimator is implemented, the deployed pipeline does not deconvolve, which is exactly why motion blur shows the smallest detection gain in Table XI.

2) *Low-light restoration and the ordering result*: One experiment justifies the pipeline’s denoise-before-brighten ordering. On night frames (8.60 dB, SSIM 0.247 degraded), auto-gamma gives 16.76 dB, global histogram equalisation 17.57 dB, CLAHE 12.38 dB and gamma+CLAHE 15.92 dB.

Crucially, denoising *before* enhancement yields **18.37 dB** and SSIM **0.658**, while exactly the same two operations in the opposite order yield 16.15 dB and 0.526. The 2.2 dB and 0.13 SSIM difference comes from nothing but ordering. The reason is that the enhancement applies a gain of roughly $4\times$ to a dark frame, and that gain multiplies the noise as surely as the signal; once amplified and clipped, the noise cannot be removed as cleanly. This result is what the gain-aware trigger in Fig. 2 implements: the pipeline tests $\hat{\sigma} \cdot \gamma$ so that a dark, mildly noisy frame is denoised *before* it is brightened.

TABLE VIII

SEGMENTATION INSIDE PERSON BOXES (100 PENN-FUDAN IMAGES).
IoU/DICE AGAINST PIXEL MASKS; TIME IS PER FRAME FOR ALL PERSONS.

Method	Family	GT boxes		YOLO boxes	
		IoU	Dice	IoU	ms
box (baseline)	—	0.497	0.658	0.473	0.1
Canny + morph.	edge	0.286	0.409	0.276	4.3
Sobel + Otsu	edge	0.201	0.306	0.179	4.3
Otsu	region	0.426	0.579	0.404	6.2
region growing	region	0.544	0.693	0.526	838
watershed	region	0.692	0.814	0.632	3.5
GrabCut	region	0.642	0.766	0.588	401

3) *Reconstruction of lost blocks*: Diffusion-based in-painting recovers packet loss almost completely. At 3 % block loss, Telea in-painting raises PSNR from 21.60 to 35.00 dB (+13.4 dB) in 5.5 ms; at 1 % loss it reaches 40.38 dB and at 8 % still 30.39 dB. Navier–Stokes in-painting is statistically indistinguishable (34.95 dB at 3 %), so the cheaper Telea variant is the default. The naive alternative of median filtering the damaged frame is not merely worse but actively harmful (20.08 dB at 3 % loss, below the 21.60 dB degraded input), because it smears the black blocks outward while also destroying undamaged detail. The lesson generalises: reconstruction requires knowing *which* pixels are missing, which is precisely what the lost-block diagnostic provides at 100 % detection (Table IV).

E. Task 4: Segmentation

Table VIII compares both families inside the same ROIs, evaluated against Penn-Fudan’s pixel masks, first with oracle ground-truth boxes and then with the YOLO boxes the deployed system actually produces.

Three conclusions follow. First, *region-based methods dominate edge-based ones* on cluttered streets (watershed 0.692 versus Canny 0.286): pedestrian boundaries are often low-contrast, so an edge map is incomplete where contrast is weak and over-complete where the background is textured, and morphological closing cannot distinguish the two. Second, the best method is also the cheapest competitive one—watershed is 115× faster than GrabCut *and* more accurate, because a well-placed marker pair encodes the same prior GrabCut must infer by iterating colour mixtures—so it is the default in the **balanced** and **quality** profiles. Third, only watershed and region growing beat the trivial “fill the box” baseline of 0.497. Replacing oracle with YOLO boxes costs a uniform 0.04–0.06 IoU, so localisation error propagates but does not change the ranking.

1) *Does restoration help segmentation?*: Not consistently—a negative result worth stating. Under $\sigma=30$ noise, watershed IoU falls from 0.641 (clean) to 0.541; **balanced** restoration recovers only to 0.560 and **quality** (NLM) makes it *worse* at 0.513. Canny degrades monotonically with denoising strength (0.302, 0.263, 0.240, 0.203). A strong denoiser removes the weak gradients both the edge detector and the watershed transform rely on, and the noise it removed was not what limited them. Restoration

TABLE IX

QUALITY AT FIXED RATE (24 IMAGES), BY LINEAR INTERPOLATION OF EACH MEASURED RATE–DISTORTION CURVE.

Codec	0.5 bpp		1.0 bpp	
	PSNR	SSIM	PSNR	SSIM
JPEG (libjpeg)	27.01	0.801	30.69	0.902
DCT codec (ours)	25.80	0.761	29.07	0.863
Wavelet bior4.4	25.19	0.698	29.02	0.824
Wavelet Haar	24.42	0.663	27.96	0.795

is justified by its effect on *detection* (Section VI-G), not on segmentation.

2) *Motion segmentation on real CCTV*: On CAVIAR, background subtraction reaches F_1 0.457–0.475 (MOG2) and 0.475–0.554 (running average) at 0.9–1.3 ms per frame. MOG2 is more precise (0.82–0.97) but less sensitive (recall 0.13–0.32), because its per-pixel mixture absorbs a person who pauses into the background; the running average, updated only where the pixel is currently background, retains such targets longer. Both are an order of magnitude cheaper than a detector, and supply the **fast** profile’s segmentation.

F. Task 5: Compression

1) *Rate–distortion*: Table IX compares the production codec, our from-scratch DCT codec and two wavelet codecs at matched bit rates, from measured byte counts.

Our DCT codec trails libjpeg by 1.6 dB at 1 bpp despite an identical quantiser and IJG scaling. The difference is the entropy coder alone: libjpeg uses optimised Huffman tables over run-length symbols, ours a generic zlib stream over coefficient planes. Landing within 1.6 dB of the production codec validates the transform and quantisation stages. Among wavelets, CDF 9/7 beats Haar by 0.8–1.1 dB at equal rate; at 1 bpp it matches our DCT codec in PSNR but its SSIM is markedly lower (0.824 versus 0.863), reflecting diffuse ringing rather than block edges.

2) *Task-aware fidelity: how far can we compress?*: PSNR is not the operational criterion; detection is. Re-running the detector on JPEG-compressed frames gives pedestrian F_1 of 0.888 (raw, 24 bpp), 0.878 ($q=70$, 1.65 bpp), **0.884** ($q=40$, 0.95 bpp), 0.852 ($q=20$), 0.778 ($q=10$) and 0.406 ($q=5$). Detection is therefore *unharmful* down to $q=40$ —a 25× reduction in data—and degrades gracefully until it collapses below $q=10$. A surveillance network can consequently transmit at roughly 1 bpp without measurable analytic loss, which is a far more useful operating rule than any fixed PSNR target.

3) *Region-of-interest coding*: The value of ROI coding depends entirely on how much of the frame the objects occupy. On Penn-Fudan, whose pedestrians dominate the frame, ROI coding gains only 0.5 dB on people (38.63 versus 38.14 dB) while sacrificing 11 dB of background quality—a poor trade. On the wide CCTV view of `vtest.avi`, where detected objects cover just **4.5 %** of the pixels, the same scheme yields **35.06 dB on objects against 28.04 dB for uniform JPEG at the same file size**, a gain of **7.0 dB** for 2.9 dB of background sacrifice.

TABLE X
DETECTOR BENCHMARK, 170 PENN-FUDAN IMAGES, 423 INSTANCES.

Detector	P	R	F_1	AP ₅₀	ms
HOG + linear SVM	0.666	0.508	0.576	0.427	5.8
YOLOv4-tiny @320	0.838	0.941	0.886	0.919	11.6
YOLOv4-tiny @416	0.855	0.960	0.904	0.926	16.1
YOLOv4-tiny @608	0.724	0.917	0.809	0.807	27.8

TABLE XI
PEDESTRIAN F_1 WITH AND WITHOUT ADAPTIVE RESTORATION (30 IMAGES, SAME DETECTOR THROUGHOUT).

Scenario	raw	+ balanced	+ quality
clean	0.900	0.917	0.906
Gaussian $\sigma=15$	0.870	0.902	0.882
Gaussian $\sigma=30$	0.803	0.828	0.867
salt & pepper 5 %	0.766	0.900	0.911
salt & pepper 20 %	0.624	0.917	0.910
periodic	0.634	0.885	0.890
motion blur	0.706	0.750	0.706
low light	0.831	0.855	0.811
uneven light	0.893	0.910	0.892
block loss	0.911	0.917	0.899
night compound	0.509	0.722	0.723

This is the compression–fidelity trade-off resolved per region rather than per frame, and it is only available because the recognition stage supplies the object map—a direct, quantified module dependency (Section VII). It also explains why the naïve experiment on a close-up dataset would have concluded, wrongly, that ROI coding is not worth implementing.

G. Task 6: Object recognition

1) *Detector benchmark*: Table X benchmarks the detectors on all 170 Penn-Fudan images (423 labelled instances). YOLOv4-tiny at 416px is the best operating point on every metric, and the classical HOG+SVM baseline is far behind (F_1 0.576 versus 0.904).

The non-monotonic behaviour with input size is notable: enlarging the input from 416 to 608px *reduces* F_1 from 0.904 to 0.809. Recall barely moves (0.960 to 0.917) while precision collapses (0.855 to 0.724) as false positives more than double (69 to 148). At 608px the receptive field relative to object size falls below the scales the COCO anchors were tuned for, so background texture begins to fire the person class. For a resource-constrained deployment the lesson is convenient: the most expensive setting is not merely poor value, it is actively worse.

2) *Robustness and the value of restoration*: Table XI is the central interdependence experiment: identical detector, identical images, with and without the adaptive restoration stage.

Restoration never harms the clean stream and recovers most of the loss on every fault, with the largest gains exactly where the diagnostics fire hardest: impulse noise (+0.293 F_1), periodic interference (+0.256) and compound night (+0.214). Motion blur gains least (+0.044) because the pipeline does not deconvolve—the PSF is unknown at run time. On the KITTI clip, vehicle F_1 improves from 0.614 to 0.783 (Gaussian),

TABLE XII
DETECTION ON REAL CCTV (CAVIAR). F_1 TRACKS PEDESTRIAN PIXEL HEIGHT, NOT SEQUENCE DIFFICULTY IN ANY OTHER SENSE.

Sequence	median height	F_1		
		HOG	@416	@608
EnterExitCrossingPaths1cor	65 px	0.129	0.943	0.930
OneLeaveShop1cor	61 px	0.038	0.820	0.798
Meet_Crowd	26 px	0.000	0.017	0.050
Fight_Chase	21 px	0.000	0.013	0.050
Walk1	18 px	0.000	0.062	0.097

TABLE XIII
END-TO-END PERFORMANCE, 768×576, SINGLE PROCESS, FULL PIPELINE INCLUDING PRIVACY AND ROI CODING.

Stream	Profile	FPS	ms	restore	detect
clean	fast	45.5	22.0	2.4	4.0
	balanced	20.8	48.2	18.8	14.3
	quality	12.8	78.1	35.7	26.4
night+noise	fast	44.5	22.5	2.8	3.7
	balanced	18.7	53.6	21.5	15.0
	quality	2.8	360.4	276.4	29.5

0.716 to 0.826 (impulse) and 0.567 to 0.809 (periodic), so the effect is not specific to pedestrians.

3) *Real CCTV: where the system actually fails*: Table XII evaluates the detectors on five CAVIAR sequences of genuine surveillance footage. The spread is extreme and instructive.

F_1 correlates almost perfectly with median pedestrian height. The two corridor views, where people occupy 61–65px, yield 0.943 and 0.820; the three wide-angle lobby views, where the median person is 18–26px, collapse to 0.013–0.097 for every detector. This is a scale limit, not a noise or lighting limit: no restoration recovers a 20px person, because YOLOv4-tiny’s coarsest stride is comparable to the whole object. It is the most important practical finding here, because it is invisible on Penn-Fudan—whose pedestrians are hundreds of pixels tall—and would have been missed by evaluating only on curated stills. The consequence is that camera placement dominates algorithm choice: pedestrians must subtend roughly 40–50px, or tiled per-tile inference must be added. HOG+SVM fails on all five sequences and is not viable for real CCTV.

4) *Privacy versus accessibility*: Anonymising heads preserves the analytics: F_1 falls only from 0.900 to 0.877 (pixelation) or 0.872 (blurring), so counting and tracking still work on the public stream, whereas pixelating whole bodies destroys them (F_1 0.026). Identity and detectability are separable only if the protected region is chosen carefully. The keyed scrambler restores the original bit-exactly for a key holder and yields an unrecoverable result under the wrong key.

H. Real-time performance and resources

Table XIII reports the end-to-end throughput of all three profiles on 768×576 video, on both a clean stream and a degraded night stream that forces the restoration branch to execute.

All three profiles exceed real time on a clean stream, and fast and balanced remain real-time under degradation.

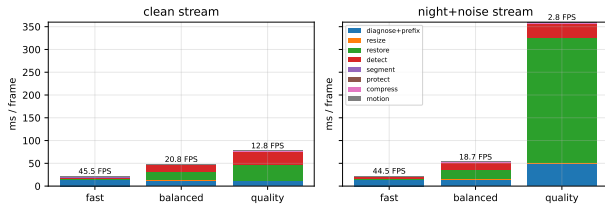


Fig. 4. Per-stage latency. On the clean stream in the fast profile, blind diagnosis (13.5 ms) is the dominant cost, not restoration or detection.

quality is the interesting case: 78 ms on a clean frame but 360 ms once the night stream triggers non-local means, which alone costs 276 ms. This is exactly what the adaptive design intends—expensive operators are paid for only when the frame warrants them—but it shows a worst-case latency budget must be sized for the degraded case.

Fig. 4 exposes an unanticipated cost: in the fast profile on a clean stream the diagnostic front end consumes 13.5 of the 22.0 ms budget—more than detection and restoration combined. Diagnosis is what allows every other stage to be skipped, so it is paid on every frame while the stages it gates are not; amortising it over N frames is the most valuable remaining optimisation.

Detector input size is the principal speed control, spanning 7.6 ms (224 px, 132 FPS) to 30.1 ms (608 px, 33 FPS); with Table X, 416 px is both the most accurate and a mid-cost setting, so the accuracy/speed trade-off is unusually benign. Peak resident memory is 1292 MB including the network and the OpenCV runtime, fitting a Jetson- or Pi-5-class device.

VII. MODULE INTERDEPENDENCE

The interdependence of the modules deserves explicit analysis. Fig. 5 shows the dependency graph, and this project quantifies each edge rather than asserting it.

Restoration $\rightarrow$ *recognition* is the strongest coupling. The same detector on the same scenes gains +0.293 F_1 under impulse noise, +0.256 under periodic interference and +0.214 on compound night frames (Table XI). Restoration is not cosmetic; it is what makes the detector usable on degraded input.

Recognition $\rightarrow$ *segmentation* is a hard dependency: every single-image segmenter runs inside a detector box, and substituting YOLO for oracle boxes costs 0.04–0.06 IoU, so localisation error propagates but does not dominate.

Recognition $\rightarrow$ *compression* is worth the most bits: detection-driven ROI coding yields +7.0 dB on objects at equal file size (Section VI-F), because the codec can only protect regions the detector identifies.

Recognition $\rightarrow$ *privacy* determines what is hidden. Since the head region derives from the detection box, a missed detection is also a privacy failure—an availability requirement becomes a confidentiality one, worth stating explicitly in a security review.

Restoration $\rightarrow$ *segmentation* is the coupling that does not pay (Section VI-E); segmentation consumes the restored frame

anyway because the detector needs it, so no un-restored path is kept.

VIII. DISCUSSION OF THE KEY TRADE-OFFS

We close by answering each of the five key requirements with a measurement.

Real-time versus quality. The profiles span 45.5 FPS (fast) to 12.8 FPS (quality) for a detection difference of only 0.011 F_1 on clean frames but up to 0.064 on degraded ones. The correct policy is therefore conditional—run fast until the diagnostics report degradation, then escalate—and the diagnostics needed to implement it already exist.

Resources versus performance. Detector input size spans 4 $\times$ in throughput and the accuracy optimum (416 px) is not the most expensive setting. Peak memory of 1292 MB fits an edge device; no GPU is used.

Noise removal versus detail. The adaptive median preserves detail a fixed median destroys, and the gain-aware trigger avoids denoising frames that do not need it. That over-filtering is real is clearest in segmentation: NLM lowers watershed IoU from 0.560 to 0.513 against the cheaper bilateral filter.

Compression versus fidelity. Pedestrian F_1 is unchanged to JPEG $q=40$ (0.95 bpp, 25 $\times$) and collapses below $q=10$. Fidelity the analytics cannot perceive is worth discarding, and ROI coding discards it only in the background.

Privacy versus accessibility. Head anonymisation costs 0.023 F_1 while removing identity; whole-body anonymisation costs 0.874 and is unusable. A keyed scrambler preserves lawful access and an HMAC tag makes tampering detectable, serving three audiences—public, operator, investigator—from one video store.

IX. LIMITATIONS AND FUTURE WORK

Small objects dominate the error budget. Detection fails below roughly 30 px of pedestrian height regardless of restoration (Table XII). Tiled inference, a higher-capacity detector or—most cheaply—correct camera siting are the remedies. This is a property of the detector and the geometry, not of the image-processing pipeline.

Uneven illumination is not blindly detectable (Section VI-A), so homomorphic correction is exposed as a per-camera installation setting rather than an automatic decision.

Deblurring assumes a known PSF. Wiener restoration loses several dB when the blur length is misestimated, so blind PSF estimation is the natural next step; it is also why motion blur shows the smallest detection gain.

The noise estimator under-reads heavy noise because of clipping, so thresholds were calibrated against its measured output rather than nominal σ .

The keyed scrambler is a teaching construction. Block permutation with a key-derived XOR keystream demonstrates reversible masking but is not an authenticated cipher; a deployment should use AES-GCM with vaulted keys.

Synthetic degradation is required because reference metrics need a clean ground truth that real night footage cannot provide; the CAVIAR and KITTI evaluations partially offset this with genuine footage.

Module interdependence — every downstream task consumes an upstream product

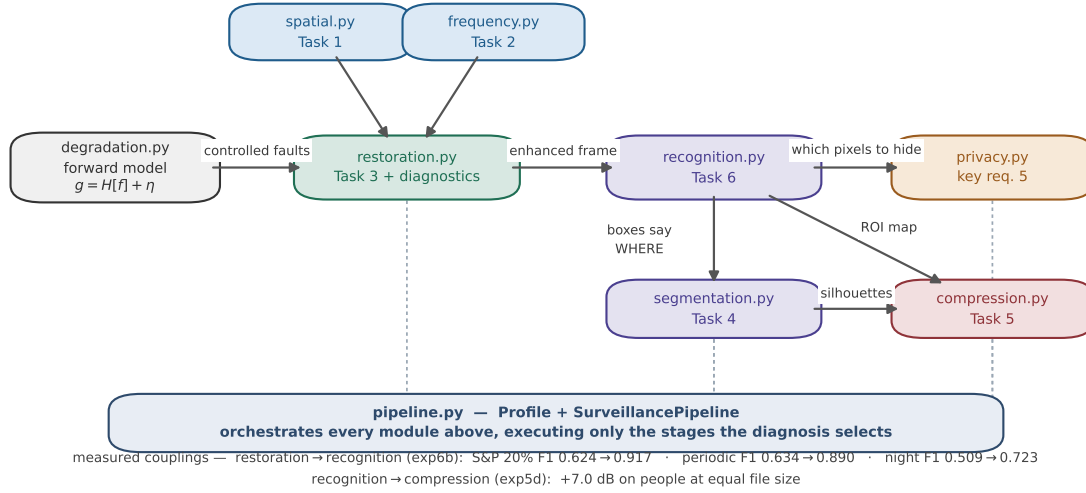


Fig. 5. Module interdependence. Solid arrows are data dependencies; the dashed ties indicate that `pipeline.py` orchestrates each module. The captions carry the measured value of each coupling.

X. CONCLUSION

We presented an adaptive image-processing pipeline for smart-city surveillance that diagnoses each frame blindly and executes only the restoration stages the diagnosis justifies, under a profile capping the cost of each stage. All six technical tasks are implemented, with the core operators written from first principles and verified against reference libraries.

The measured outcomes: blind detection of five of six fault classes at 100 % with at most 5 % false alarms; restoration lifting pedestrian F_1 from 0.624 to 0.917 under heavy impulse noise and 0.509 to 0.723 on compound night frames; watershed segmentation at IoU 0.692 in 3.5 ms, beating GrabCut on both accuracy and cost; ROI coding worth +7.0 dB on objects at equal file size; and head anonymisation that removes identity for 0.023 F_1 . The system runs at 45.5 FPS (fast) and 20.8 FPS (balanced) on one CPU within 1.3 GB.

Two negative results are as informative as the positive ones. Denoising does not improve segmentation, because strong denoisers remove the weak gradients it depends on. And detection on genuine CCTV is governed by object scale rather than image quality: below about 30px of pedestrian height every detector fails regardless of restoration. The most valuable future work is therefore not a better filter but blind PSF estimation, amortised diagnostics and tiled inference for small objects.

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